

UNIT 3.1 - Mathematical induction

Consider the statement

$$\frac{d}{dx}x^n = nx^{n-1}.$$

This is supposed to be true for all positive integers n .

But we have not yet proved this for all n .

We noticed that it is true for $n = 1, 2$ and 3 , and then took it for granted that it will hold for all positive integers n .

Of course this can be dangerous.

There is a statement in terms of prime numbers which is true for all integers from 1 through 40 , but fails to hold for $n = 41$.

On the other hand, one does not live long enough to test a statement for ALL positive integers.

Mathematical Induction is a very powerful instrument with which one can prove a statement which is true for all positive integers $n = 1, 2, 3, \dots$

In this study unit we shall state the principle of mathematical induction (Notes, 3.1 on p 60) and then you have to prove a number of statements for all positive integers n .

In this connection we would like to make a number of comments.

Firstly, in these proofs one assumes the given statement to be true for $n = k$. Why can one do this?

Secondly, be careful never to assume that which is to be proved.

Thirdly, try to be really honest when proving the statement for $n = k + 1$.

After having cheated somewhere along the line, one has to cheat again to end up with that which is to be shown. So rather don't cheat!

Carefully work through the example 1 in the notes and then try the exercises. The solutions to some of these exercises are given in Solutions.